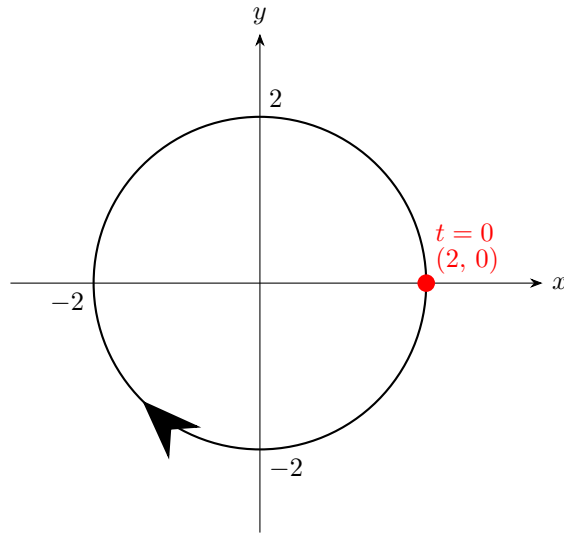


Week 3 Tutorial Solutions

Question 11

(a)



Okay so remember to draw out your trig graphs which include $\cos(t)$, $\sin(t)$, $-\cos(t)$ and $-\sin(t)$.

The red dot in our graph above represents the starting point at time $t = 0$. We can use this starting point to construct the x -component and y -component of our vector functions.

- $x(0)$: x starts at a maximum and decreases initially $\Rightarrow \cos(t)$
- $y(0)$: y starts at 0 and decreases initially $\Rightarrow -\sin(t)$
- centered at $(a, b) = (0, 0)$
- $r = 2$
- Speed = 3 m/s

$$\omega = \frac{v}{r} = \frac{3}{2} \text{ rad/s}$$

$$\mathbf{r}(t) = \langle r \cos(\omega t) + a, r \sin(\omega t) + b \rangle$$

$$\boxed{\mathbf{r}(t) = \langle 2 \cos\left(\frac{3}{2}t\right), -2 \sin\left(\frac{3}{2}t\right) \rangle}$$

(b)

Whenever a problem like this mentions ‘speed’, we have to think about the magnitude of the velocity vector function. Why? Velocity is composed of magnitude and direction, so if we compute the magnitude of the vector function, we obtain its speed.

$$\begin{aligned}\mathbf{v}(t) = \mathbf{r}'(t) &= \left\langle -2 \sin\left(\frac{3}{2}t\right) \cdot \frac{3}{2}, -2 \cos\left(\frac{3}{2}t\right) \cdot \frac{3}{2} \right\rangle \\ &= \left\langle -3 \sin\left(\frac{3}{2}t\right), -3 \cos\left(\frac{3}{2}t\right) \right\rangle\end{aligned}$$

$$\begin{aligned}\text{Speed} = \|\mathbf{v}(t)\| &= \sqrt{\left(-3 \sin\left(\frac{3}{2}t\right)\right)^2 + \left(-3 \cos\left(\frac{3}{2}t\right)\right)^2} \\ &= \sqrt{9 \sin^2\left(\frac{3}{2}t\right) + 9 \cos^2\left(\frac{3}{2}t\right)} \\ &= \sqrt{9 \left(\sin^2\left(\frac{3}{2}t\right) + \cos^2\left(\frac{3}{2}t\right)\right)} \\ &= \sqrt{9(1)} \\ &= \boxed{3 \text{ m/s}}\end{aligned}$$

(c)

There are two ways to get an equation with only t . The first is to use the fast method of setting up the ratio between the two components. This is shown as:

Method 1: Setup parallel condition

$$\begin{aligned}\mathbf{v}(t) &= k \langle -2, 2 \rangle \quad (k \neq 0) \\ \left\langle \underbrace{-3 \sin\left(\frac{3}{2}t\right)}_{x_1}, \underbrace{-3 \cos\left(\frac{3}{2}t\right)}_{y_1} \right\rangle &= \left\langle \underbrace{-2k}_{x_1}, \underbrace{2k}_{y_1} \right\rangle\end{aligned}$$

Equating the components gives our x - and y -equations:

$$x: \quad -3 \sin\left(\frac{3}{2}t\right) = -2k \tag{1}$$

$$y: \quad -3 \cos\left(\frac{3}{2}t\right) = 2k \tag{2}$$

Divide (1) by (2):

$$\begin{aligned}\frac{-3 \sin\left(\frac{3}{2}t\right)}{-3 \cos\left(\frac{3}{2}t\right)} &= \frac{-2k}{2k} \\ \tan\left(\frac{3}{2}t\right) &= -1\end{aligned}$$

Method 2: Substitution

Another approach we can take is essentially the same thing but using the standard steps taught in class.

Setup parallel condition again:

$$\begin{aligned}\mathbf{v}(t) &= k \langle -2, 2 \rangle \quad (k \neq 0) \\ \left\langle \underbrace{-3 \sin\left(\frac{3}{2}t\right)}_{x_1}, \underbrace{-3 \cos\left(\frac{3}{2}t\right)}_{y_1} \right\rangle &= \left\langle \underbrace{-2k}_{x_1}, \underbrace{2k}_{y_1} \right\rangle\end{aligned}$$

This gives the same x - and y -equations, (1) and (2).

Isolate k in the x -equation (1):

$$\begin{aligned} -3 \sin\left(\frac{3}{2}t\right) &= -2k \\ k &= \frac{3}{2} \sin\left(\frac{3}{2}t\right) \end{aligned} \quad (3)$$

Substitute our equation for k , (3), into the y -equation (2):

$$\begin{aligned} -3 \cos\left(\frac{3}{2}t\right) &= 2k \\ -3 \cos\left(\frac{3}{2}t\right) &= 2 \left(\frac{3}{2} \sin\left(\frac{3}{2}t\right) \right) \\ -3 \cos\left(\frac{3}{2}t\right) &= 3 \sin\left(\frac{3}{2}t\right) \\ -\cos\left(\frac{3}{2}t\right) &= \sin\left(\frac{3}{2}t\right) \\ -1 &= \frac{\sin\left(\frac{3}{2}t\right)}{\cos\left(\frac{3}{2}t\right)} \\ -1 &= \tan\left(\frac{3}{2}t\right) \end{aligned}$$

Both methods lead to the same equation:

$$\tan\left(\frac{3}{2}t\right) = -1$$

Solving with the CAST rule

The CAST rule

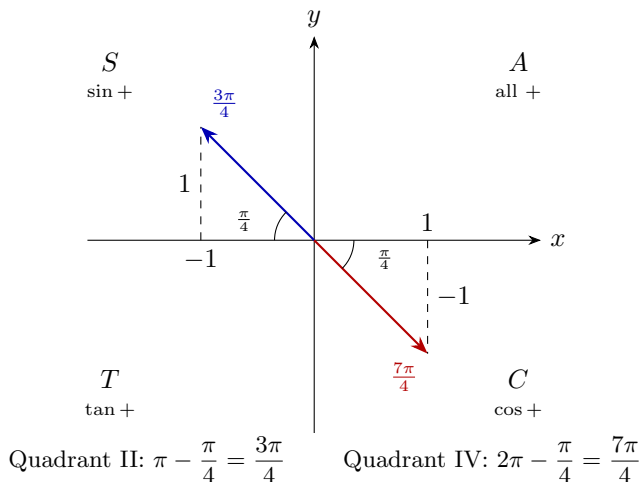
The CAST rule tells us which trig ratios are *positive* in each quadrant. Starting in quadrant IV and moving counterclockwise, the letters spell C-A-S-T:

- **A** (Quadrant I): **A**ll of sin, cos, tan are positive.
- **S** (Quadrant II): only **S**ine is positive.
- **T** (Quadrant III): only **T**angent is positive.
- **C** (Quadrant IV): only **C**osine is positive.

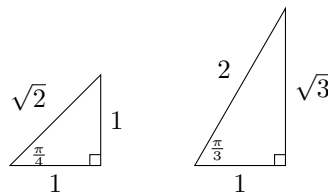
To use it, first find the *reference angle* θ_{ref} (the acute angle between the terminal arm and the x -axis) using the special triangles. Then use CAST to decide which quadrants the angle can be in.

Since $\tan\left(\frac{3}{2}t\right) = -1$ is *negative*, the angle $\frac{3}{2}t$ must be in the quadrants where tangent is *not* positive: **S** (Quadrant II) and **C** (Quadrant IV).

Reference angle: $\tan \theta_{\text{ref}} = \frac{\text{opp}}{\text{adj}} = \frac{1}{1} = 1 \Rightarrow \theta_{\text{ref}} = \frac{\pi}{4}$ (from the $\frac{\pi}{4}$ special triangle).



Special Triangles



Taking the Quadrant II angle:

$$\frac{3}{2}t = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$t = \frac{6\pi}{12}$$

$$t = \boxed{\frac{\pi}{2}}$$

Taking the Quadrant IV angle:

$$\frac{3}{2}t = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$

$$t = \frac{14\pi}{12}$$

$$t = \frac{7\pi}{6}$$

Extra Insights into the Solution (this part is NOT necessary to cover for this tutorial question, this is just for your own understanding and curiosity)

How Do We Get multiple solutions for t ?

The particle keeps going around the circle, so its velocity is parallel to $\langle -2, 2 \rangle$ more than once. The CAST diagram gives us *two* angles in one rotation, $\frac{3\pi}{4}$ (Quadrant II) and $\frac{7\pi}{4}$ (Quadrant IV), and each one repeats every full rotation of 2π :

$$\text{Quadrant II: } \frac{3}{2}t = \frac{3\pi}{4} + 2\pi n \quad \implies \quad t = \frac{\pi}{2} + \frac{4\pi}{3}n$$

$$\text{Quadrant IV: } \frac{3}{2}t = \frac{7\pi}{4} + 2\pi n \quad \implies \quad t = \frac{7\pi}{6} + \frac{4\pi}{3}n$$

where $n = 0, 1, 2, \dots$ (so that $t \geq 0$). The first few solutions are:

n	Quadrant II	Quadrant IV
0	$t = \frac{\pi}{2}$	$t = \frac{7\pi}{6}$
1	$t = \frac{11\pi}{6}$	$t = \frac{5\pi}{2}$

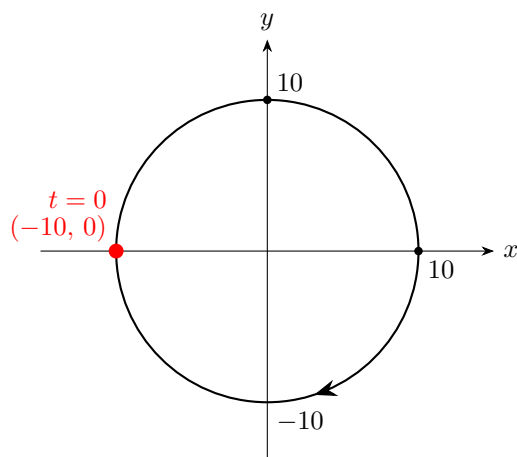
Since the two quadrants are exactly π apart, both families combine into one formula:

$$\frac{3}{2}t = \frac{3\pi}{4} + \pi n \quad \implies \quad t = \frac{\pi}{2} + \frac{2\pi}{3}n, \quad n = 0, 1, 2, \dots$$

Note on direction. From (3), $k = \frac{3}{2}\sin(\frac{3}{2}t)$. In Quadrant II, $\sin > 0$, so $k > 0$ and the particle moves in the *same* direction as $\langle -2, 2 \rangle$. In Quadrant IV, $\sin < 0$, so $k < 0$ and the particle moves in the *opposite* direction. Both count as parallel because $k \neq 0$. For example, at $t = \frac{7\pi}{6}$:

$$\mathbf{v}\left(\frac{7\pi}{6}\right) = \left\langle \frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2} \right\rangle = -\frac{3\sqrt{2}}{4} \langle -2, 2 \rangle$$

Question 12



This uses the same ideas from question 11 in terms of creating x -components and y -components from our trig graphs. Let's see how it works here.

- starts at $(-10, 0)$ (red dot)
- centered at $(a, b) = (0, 0)$
- $r = 10$
- $v = 5$ m/s

$$\omega = \frac{5}{10} = \frac{1}{2} \text{ rad/s}$$

For time $t = 0$:

- $x(0)$: starts at a minimum and increases initially $\Rightarrow -\cos(t)$
- $y(0)$: starts at 0 and increases initially $\Rightarrow \sin(t)$

$$\mathbf{r}(t) = \langle r \cos(\omega t) + a, r \sin(\omega t) + b \rangle$$

$$\boxed{\mathbf{r}(t) = \langle -10 \cos(\tfrac{1}{2}t), 10 \sin(\tfrac{1}{2}t) \rangle}$$

Question 13

(a)

$$\mathbf{r}(t) = \langle t - 2, t^2 + 1 \rangle$$

We want to sketch this on the xy -plane, so we eliminate the parameter (find an equation of the form $y = f(x)$). Write out the x - and y -equations:

$$x = t - 2 \tag{4}$$

$$y = t^2 + 1 \tag{5}$$

Solve the x -equation (4) for t in terms of x :

$$t = x + 2 \tag{6}$$

Substitute (6) into the y -equation (5):

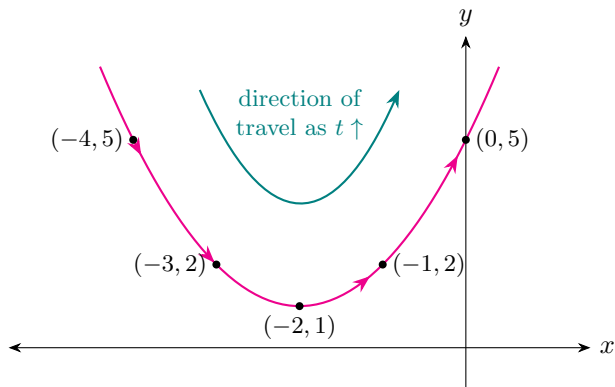
$$y = (x + 2)^2 + 1 = f(x)$$

Plot: $y = (x + 2)^2 + 1$

Set $x = 0$ to get the y -intercept:

$$y = (0 + 2)^2 + 1 = 5$$

Draw the vertex which is at $(-2, 1)$ and any other common points you find by substituting x values into the function.



(b)

Position and velocity vector (tangent vector) at $t = -1$.

Small Note: remember that the derivative of a function at a point equals the slope of the tangent line to the curve at that exact point. This means our tangent line at a specific point on the position function curve will be the velocity at that point.

$$\mathbf{r}(t) = \langle t - 2, t^2 + 1 \rangle$$

Substitute $t = -1$:

$$\begin{aligned} \mathbf{r}(-1) &= \langle -1 - 2, (-1)^2 + 1 \rangle \\ \therefore \mathbf{r}(-1) &= \boxed{\langle -3, 2 \rangle} \end{aligned}$$

Tangent vector (velocity) at $t = -1$:

$$\mathbf{r}'(t) = \langle 1, 2t \rangle$$

$$\mathbf{r}'(t) = \mathbf{v}(t)$$

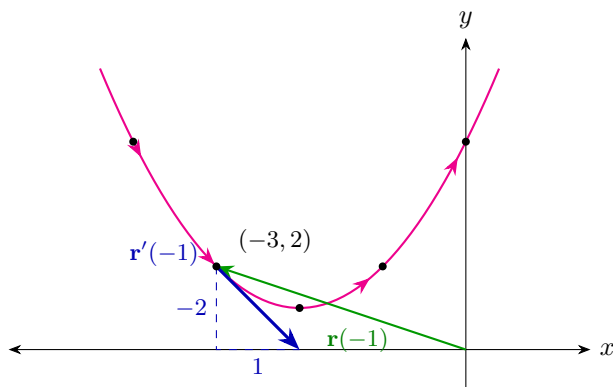
$$\mathbf{v}(-1) = \langle 1, 2(-1) \rangle$$

$$\therefore \mathbf{v}(-1) = \boxed{\langle 1, -2 \rangle}$$

Our position vector is drawn from the origin. The velocity vector is drawn with its tail on the curve at $(-3, 2)$. It represents the direction and magnitude of the rate of change of $\mathbf{r}(t)$ at $t = -1$.

Question: Why is the position vector drawn from the origin and the velocity vector drawn from the point $\mathbf{r}(-1)$?

Answer: The position vector tells us where the particle is relative to the origin. The velocity vector tells us how the particle is moving at that location, so we place its tail at the particle and point it in the direction of motion.



Problem 14

Tenderheart Bear, one of the mystical Care Bears™, is driving in his Cloud Car, with his position given by

$$\mathbf{r}(t) = \langle t + 1, t^2 + 2 \rangle, \quad -\infty < t < \infty$$

His friend Funshine Bear is standing at coordinates $(-1, 2)$.

When Tenderheart presses a button in his Cloud Car, the car will send a magical ‘hug’ straight ahead in the direction the car is moving at that moment. If Tenderheart wants to send a hug to Funshine without having to stop, at what time t should Tenderheart press the button?

Define the time that the ‘hug’ is released. Let $t = a$ be the release time.

We need to find a straight line that is:

1. tangent to the curve $\mathbf{r}(t)$ at time $t = a$;
2. intersects the point $(-1, 2)$ at a time t , with $t > a$.

Projectile motion

A ball released from position x_0 at time t_0 has x -direction position described by

$$x(t) = v_0 (t - t_0) + x_0,$$

where v_0 is the velocity at t_0 .

We can define our line as:

$$\mathbf{L}(t) = \underbrace{\mathbf{v}(a)}_{\substack{\text{velocity of the bear} \\ \text{at the time of release} \\ \text{(slope of } \mathbf{L}(t))}} \underbrace{(t - a)}_{\substack{\text{elapsed time} \\ \text{since release}}} + \underbrace{\mathbf{r}(a)}_{\substack{\text{initial position at} \\ \text{point of release}}} \quad (7)$$

Note that when $t = a$, $\mathbf{L}(a)$ must equal $\mathbf{r}(a)$.

$$\mathbf{r}(t) = \left\langle \overbrace{t+1}^x, \overbrace{t^2+2}^y \right\rangle$$

Find $\mathbf{v}(t)$ (differentiate the x and y components):

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle 1, 2t \rangle$$

Substitute $\mathbf{v}(a)$ and $\mathbf{r}(a)$ into $\mathbf{L}(t)$ (7):

$$\mathbf{L}(t) = \underbrace{\overbrace{\langle 1, 2a \rangle}^{\mathbf{v}(a)}(t - a)}_{\substack{\text{scalar multiplication}}} + \underbrace{\overbrace{\langle a + 1, a^2 + 2 \rangle}^{\mathbf{r}(a)}}_{\substack{\text{vector} \\ \text{addition}}}$$

$$\begin{aligned} \mathbf{L}(t) &= \langle t - a, 2a(t - a) \rangle + \langle a + 1, a^2 + 2 \rangle \\ &= \langle t - a + (a + 1), 2at - 2a^2 + (a^2 + 2) \rangle \\ &= \langle t + 1, -a^2 + 2at + 2 \rangle \end{aligned}$$

Set the x and y components equal to the coordinates $(-1, 2)$ (where Funshine is):

$$x: \quad t + 1 = -1 \tag{8}$$

$$y: \quad -a^2 + 2at + 2 = 2 \tag{9}$$

Solve the x -equation (8):

$$t + 1 = -1 \implies \therefore t = -2$$

Substitute $t = -2$ into the y -equation (9):

$$-a^2 + 2a(-2) + 2 - 2 = 0$$

$$-a^2 - 4a = 0$$

$$-a(a + 4) = 0 \tag{factor}$$

There are two solutions: $a = 0$ and $a = -4$.

Recall the requirement $t > a$. Since $t = -2$, only $a = -4$ is physically valid.

Tenderheart should release the hug at time $t = -4$.
The hug will reach the point $(-1, 2)$ at $t = -2$.